

Requirements				
Req. ID	Requirement Title	Requirement Intent	Version ID	Status
WMC-REQ-001	System Initialization	On System start, the controller shall enter the Idle stage with all actuators OFF, cycle_complete set to 0, and fault_indicator set to 0.	Ver. 01	Pass
WMC-REQ-002	Start Condition	When start_button is pressed and door_closed is 1, the controller shall start the washing cycle.		Pass
WMC-REQ-003	Door Safety Before Start	If start_button is pressed while door_closed is 0, the controller shall remain in Idle state and not initiate the cycle.		Pass
WMC-REQ-004	Door Lock Activation	After a valid start command, the controller shall activate door_lock and shall wait for door_lock_delay duration before entering the Water_Fill_Wash_Cycle stage.		Pass
WMC-REQ-005	Wash-Cycle Water Fill	During Water_Fill_Wash_Cycle stage, the controller shall turn ON water_valve and shall remain in this stage until water_level_ok becomes 1. The water_level_ok is shared with the rinse-cycle water fill stage.		Pass
WMC-REQ-006	Detergent Fill Stage	After wash-cycle water filling is completed, the controller shall enter Detergent_Fill stage and turn ON detergent_valve until detergent_level_ok becomes 1.		Pass
WMC-REQ-007	Wash Stage	After detergent filling is completed, the controller shall enter the Wash stage and turn on wash_motor for the configured wash_duration.		Pass
WMC-REQ-008	Wash-Cycle Drain	After wash_duration is completed, the controller shall enter Drain_Wash_Cycle stage and turn ON drain_pump for the configured drain_duration. The drain_duration parameter is shared with the Drain_Rinse_Cycle stage.		Pass
WMC-REQ-009	Rinse-Cycle Water Fill	Water_Fill_Rinse_Cycle stage and shall turn ON water_valve until water_level_ok become 1. The water_level_ok signal is shared with the wash-cycle water fill stage.		Pass
WMC-REQ-010	Rinse Stage	After rinse-cycle water filling is completed, the controller shall enter the Rinse stage and turn ON wash_motor for the configured rinse_duration.		Pass
WMC-REQ-011	Rinse-Cycle Drain	After rinse_duration is completed, the controller shall enter Drain_Rinse_Cycle stage and turn ON drain_pump for the cofigured drain_duration. The drain_duration parameter is shared with the Drain_Wash_Cycle stage.		Pass
WMC-REQ-012	Spin Stage	After rinse-cycle draining is completed, the controller shall enter Spin stage and turn ON the spin_motor for the configured spin_duration.		Pass
WMC-REQ-013	Cycle Completion	After spin_duration is completed, the controller shall enter Cycle_Complete stage, turn OFF all actuators, shall unlock the door_lock, and shall set cycle_complete to 1.		Pass